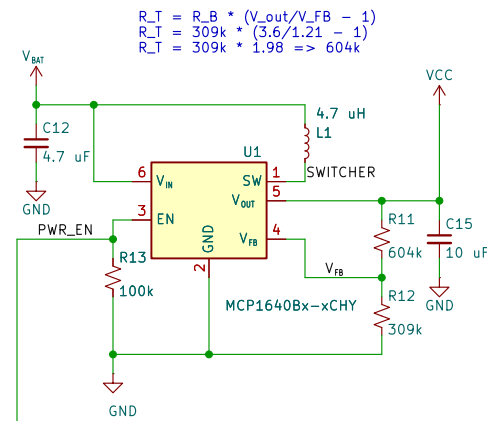
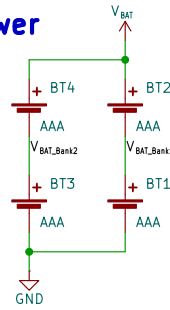
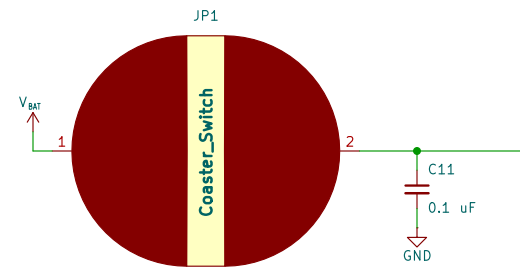


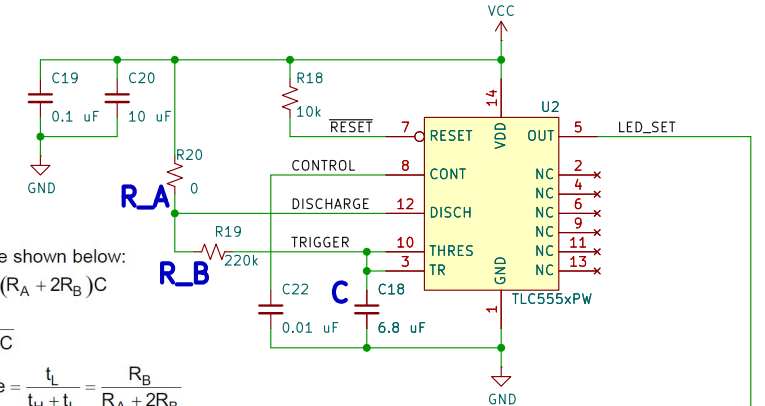
Power



Main Switch



VARIANT A



$$t_H = 0.693(R_A + R_B)C$$

$$t_L = 0.693 (R_B) C$$

Other useful relationships are shown below:

$$\text{period} = t_H + t_L = 0.693 (R_A + 2R_B) C$$

$$\text{frequency} \approx \frac{1.44}{(R_A + 2R_B)C}$$

$$\text{Output driver duty cycle} = \frac{t_L}{t_H + t_L} = \frac{R_B}{R_A + 2R_B}$$

$$\text{Output waveform duty cycle} = \frac{t_H}{t_H + t_L} = 1 - \frac{R_B}{R_A + 2R_B}$$

$$\text{Low-to-high ratio} = \frac{t_L}{t_H} = \frac{R_B}{R_A + R_B}$$

$$T_H = T_L \Rightarrow R_A = 0$$

$$R_B = 220k$$

$$C = 6.8 \text{ uF}$$

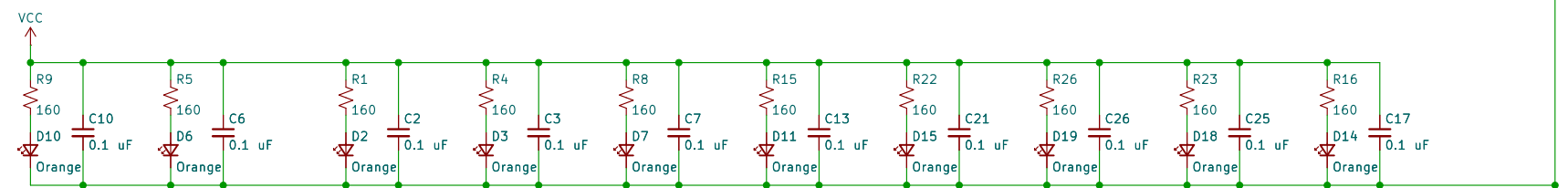
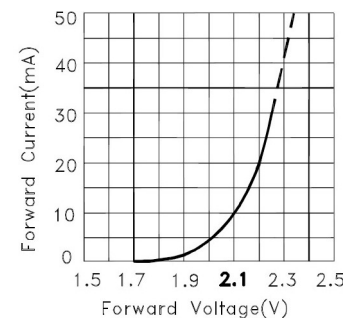
LOGOs



MAD PHYSICIST



Orange LEDs



Blue LEDs

